Priyanshu Roy

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Professional Summary

Machine Learning student at IEM Newtown targeting AI Software Engineer and Machine Learning Engineer roles. Passionate about building end-to-end AI systems, with hands-on experience designing modular RAG pipelines, stateful agentic workflows, and secure knowledge ingestion architectures for large language models (LLMs) and NLP applications. Skilled in navigating the full machine learning lifecycle—from data curation and local model optimization to cross-platform software deployment with zero-leakage privacy boundaries. Currently expanding my expertise into Reinforcement Learning to build more adaptive, self-improving agents, while remaining dedicated to bridging theoretical AI with deployable, maintainable software architecture.

Education

Institute of Engineering and Management, Kolkata — July 2024 – July 2028
Bachelor of Technology in Computer Science and Engineering (AI & ML)

Calcutta Public School, Kolkata — May 2024
Higher Secondary (ISC) - Science with Computer

Technical Skills

Languages & Scripting:
Python, Java, JavaScript, TypeScript, Rust, C/C++, Bash

Frontend & Deployment:
React, Next.js, Kotlin

Backend & Database:
Python, Node.js, Rust, PostgreSQL

Machine Learning:
PyTorch, Tensorflow, RAG, LLMs, Transformers, NLP, CNNs, ONNX, Quantization

Artificial Intelligence:
Agentic Workflow, LangChain, LangGraph

DevOps & Version Control:
Git, Docker, CI/CD

Projects

HackT – RAG Agent

Tech: Tauri (Rust + React), Python, LanceDB + KùzuDB, VLM, Quantized LLM, AI Agent

- Architected a fully offline, multi-modal AI agent with a Tauri (Rust & React) frontend orchestrating local Python microservices via WebSocket/HTTP for zero-cloud dependency
- Designed a GPU-aware model placement strategy with dynamic VRAM swapping (Florence-2 and Qwen-3.5) to operate within 8GB VRAM constraints without OOM failures
- Engineered a local cryptographic knowledge vault using hybrid vector-graph retrieval (LanceDB + KùzuDB) with chunked ingestion pipelines, enabling secure, personalized threat intelligence without data exfiltration

Avaani AI – Multimodal Conversational Agent

Tech: Next.js, Python, PostgreSQL, Agentic Workflow, VRM

- Produced a modular agentic workflow with decoupled "Ears" (Faster-Whisper STT), "Mouth" (Piper TTS), and "Vision" (CV) engines coordinated via a custom state machine for low-latency multimodal interaction
- Crafted a LangGraph-inspired orchestration layer managing conversation memory, intent routing, and fallback logic, enabling context-aware responses across voice/text/image input modalities
- Deployed as a responsive web application with REST/WS API backend, with optional PyInstaller packaging for standalone Windows execution—demonstrating architecture-agnostic deployment flexibility

GoPrivate – Native Mobile Security Engine

Tech: Python, Kotlin, PyTorch, ONNX, Transformers, GradientBoost

- Devised an edge-optimized privacy engine executing behavioral analysis of mobile applications entirely on-device, with ONNX-converted models and SHAP-based feature extraction for interpretable risk scoring
- Integrated a Kotlin ↔ Python bridge for native Android execution, ensuring zero network dependency and compliance with strict data-sovereignty requirements for sensitive user environments
- Engineered a quantization-aware inference pipeline calibrated for mobile hardware constraints, delivering real-time security insights without cloud round-trips or battery-intensive processing

EduBridge – Agentic Learning Platform

Tech: Next.js, Prisma, Inngest, Agentic Workflow

- Architected a dual-interface system (web landing page + minimal in-app UI) with Google SSO and Supabase for unified authentication, leveraging Vercel for adaptive rendering based on deployment context
- Engineered a server-side agentic orchestration layer to manage personalized learning pathways and content retrieval, utilizing PostgreSQL-backed state persistence and async task queues for real-time progress tracking
- Optimized inference performance by implementing context caching for static user data, behavioral preferences, and persistent state, significantly reducing token consumption and processing latency

Achievements

1. Highly active github codebase demonstrating consistent development activity – Solved 300+ problems on HackerRank and LeetCode, demonstrating strong problem-solving and algorithmic thinking.
2. Participated in various hackathons, contributed to open-source projects by adding new features, fixing bugs, and collaborating with the developer community.